

MURAT RAIMBEKOV

ML/CV Engineer

Bishkek, Kyrgyzstan | Open to remote

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Summary

ML Engineer with hands-on experience in computer vision and tabular ML. 26 merged PRs to Ultralytics, OpenCV, HuggingFace Transformers. Building production ML systems for real estate valuation. Strong foundation in PyTorch, gradient boosting, and feature engineering.

Experience

Junior Machine Learning Engineer

Jan 2026 – Present

Baker Tilly

Bishkek, Kyrgyzstan

- Building **Atlas** — automated real estate valuation platform for Kyrgyzstan market
- Fine-tuned **XGBoost**, **LightGBM** models achieving **R² 0.70**, **MAPE 8%** on property valuation
- Used LLM prompts to extract structured data (amenities, condition, materials) from listing descriptions

Data Science Intern

Jul 2025 – Dec 2025

Baker Tilly

Bishkek, Kyrgyzstan

- Performed data collection, cleaning, and feature engineering on **10,000+ property records**
- Built data pipelines for scraping and processing real estate listings from multiple sources

Projects

Real Estate Valuation (Kazakhstan) | *XGBoost, CatBoost, ResNet, Optuna* | [GitHub](#)

2025

- Built property valuation model for Astana market achieving **R² 0.83**, **MAPE 9%** — competitive with state-of-the-art research
- Engineered **45 features**: H3 geospatial tiles, POI distances, target encoding, GroupKFold validation
- Integrated **computer vision** module using ResNet to extract quality signals from **70,000+ property photos**
- Scraped **18,000+ listings** from krisha.kz; published dataset on Kaggle

AirVision: PM2.5 Estimation from Webcams | *ResNet-50, Multimodal ML* | [GitHub](#)

2025

- Building deep learning system to estimate air pollution levels from urban webcam images in Bishkek
- Multimodal pipeline combining **ResNet-50** visual features with meteorological data from APIs
- Integrated OpenWeatherMap, IQAir, and OpenAQ APIs for ground truth PM2.5 measurements

Open Source

26 merged PRs across Ultralytics YOLO (18), OpenCV (4), Transformers (2), timm (1), torchvision (1)
Bug fixes (OBB loss indexing, operator precedence), code optimizations, documentation improvements
Listed in Ultralytics documentation authors

Technical Skills

Languages: Python, C++, SQL (PostgreSQL)

ML/CV: PyTorch, YOLO, OpenCV, Albumentations, timm, segmentation-models-pytorch, scikit-learn, XGBoost, LightGBM, CatBoost

Tools: Git, Optuna, pandas, NumPy, SHAP, Matplotlib

Education

American University of Central Asia

Bishkek, Kyrgyzstan

Bachelor's degree in Applied Mathematics and Informatics

Sep 2022 – May 2026

- Relevant Coursework: Neural Networks and Deep Learning, Data Science and Machine Learning, Data Structures & Algorithms, Probability Theory and Statistics
- GPA: **3.3/4.0**